Class 33: Practical Project Brainstorm

Session 33 of 36 – Workshop: Choosing Your Capstone Idea

Data-driven Systems Engineering

Agenda

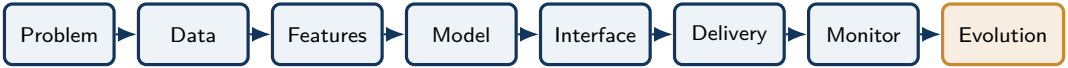
- Why planning starts with a bounded idea, not just a good model
- Defining problem, users, and scope
- Objectives you can actually evaluate
- Two project patterns from this repo, and how to propose your own
- Idea screening checklist
- Workshop deliverable: a one-page project pitch

Planning is part of the engineering work

Learning goals

- Turn a vague ML idea into a bounded problem statement.
- Distinguish MVP scope from stretch goals.
- Screen a project idea against feasibility, data availability, and evaluability.
- Leave this session with a one-page project pitch.

Course Map and Running Cases



Today in the course

Focus: Choosing and scoping a capstone project idea.
Bridge: This workshop turns the whole course arc, from Python and data-driven methods through MLOps, into the raw material for each team's own project.

Fraud case

The fraud case is used here as a worked example of a bounded, evaluable project pattern, not as a fixed assignment.

Pasteurization case

The pasteurization case is used the same way: a template for scope, not a required topic.

Course thread: A good project idea is one you can already imagine requiring, testing, deploying, and monitoring, even before you have built any of it.

Why planning matters in ML projects

- ML systems evolve through data, code, models, interfaces, validation, and operations.
- More roles means more handoffs, which means more need for shared artifacts.
- A weak idea creates downstream confusion in requirements, evaluation, and release planning.

Practical claim

Planning does not remove uncertainty, but it makes uncertainty visible enough to manage.

A project summary defines boundaries

- What problem is being solved?
- Who uses or is affected by the result?
- What is in scope for this project?
- What is intentionally out of scope?
- What evidence will convince stakeholders that the project is useful?

What a strong project summary should contain

- Problem statement and target users.
- Data sources and expected constraints.
- Expected system output: prediction, forecast, alert, or recommendation.
- Success criteria tied to technical and user outcomes.
- Risks, assumptions, and dependencies.

Objectives should be specific enough to evaluate

Weak objective

Build an ML system for quality control.

Stronger objective

Detect abnormal process trajectories from synthetic pasteurization sensor data and expose alerts through a dashboard with traceable event logs.

- Strong objectives are specific, measurable, relevant, and bounded enough to review.

Idea-generation prompts

- What data do you already have access to, or can realistically get?
- What decision would this system actually support, and for whom?
- Who would be upset, or put at risk, if this system were confidently wrong?
- Which part of the course lifecycle, data, model, interface, delivery, or monitoring, are you most curious to go deep on?

A more suitable proposal example

Project

Forecast temperature and detect abnormal state transitions in a synthetic pasteurization line.

- Data source: Flask stream from ‘pasteurization/serving.py‘.
- MVP: predict next temperature value and visualize drift alerts.
- Stretch goal: state-aware forecasting and alert triage.
- Main risks: synthetic realism, label scarcity, and operator-facing thresholds.

Another project pattern students can defend

Project

Fraud scoring as a service for transaction review support.

- Data source: credit-card fraud dataset and engineered features.
- MVP: classify transactions and expose a scoring endpoint.
- Stretch goal: threshold tuning plus analyst-facing dashboard.
- Risks: class imbalance, false-positive cost, and drift after deployment.

Or propose your own, in the same shape

- Data source: where does it come from, and can you actually access it this semester?
- MVP: the smallest version that is still a complete pipeline, not just a model.
- Stretch goal: the ambitious extension you would attempt only if the MVP lands early.
- Main risks: name at least one data risk and one operational risk.

Idea screening checklist

- Feasible: can this be built, at MVP scope, with the time and skills available?
- Data-available: is there a realistic path to the data, not just a hope for it?
- Evaluable: is there a way to say whether it worked?
- Lifecycle-complete: can you sketch, even roughly, its requirements, a test, a deployment path, and a monitoring signal?

Warning sign

If you cannot answer one of these in one sentence, the idea is not ready to pitch yet.

Common brainstorm traps

- Too big: the MVP is already a semester of work by itself.
- Too vague: “analyze the data and find insights” has no defined output.
- No data path: the idea depends on data nobody has confirmed access to.
- Pure research question: interesting, but there is no system for anyone to use.

Inspiration already sitting in this repository

Repo activity

- Notebooks and datasets used across the Python and data-driven blocks are fair game as a starting point, not only the two running cases.
- ‘documents/FraudDetectionAsaService.pdf‘ and ‘documents/Forecasting_and_Drift_Pasteurization.pdf‘ show what a scoped project write-up looks like.
- Reusing a case from class does not mean reusing the slides; it means reusing the data access and problem framing as a head start.

Discussion prompt

Question

Swap your one-page pitch with another group. What is the first question you would ask them before agreeing this project is well scoped?

- This rehearses the kind of scrutiny a proposal review will apply later.
- It also surfaces blind spots the original team stopped noticing.

Workshop deliverable: your one-page pitch

- Problem and target users, in two sentences.
- Data source and how you will access it.
- MVP scope and one stretch goal.
- Two named risks, one data-related and one operational.
- One success criterion you could check by the end of the semester.

Papers and materials

Papers and materials

Designing Machine Learning Systems; Minimum Viable Product guide; Project planning with epics and stories.

From This Class to the Next

What stays with us

- A good project idea is bounded, evaluable, has a real data path, and touches the whole lifecycle.
- MVP and stretch goals should be written down separately from the start.
- Screening an idea against a short checklist catches most scoping mistakes early.

Project use

Keep your one-page pitch; it is the direct input to the development plan built in Class 35.

Next connection

Class 34 brings in a guest speaker, Sbroiavacca, as a real-world bridge right before Class 35 turns your pitch into milestones and a development plan.

Course thread: Today's pitch is the first draft of the project that will carry through requirements, delivery, and monitoring for the rest of the course.